BRAIN DAMAGE PATIENT DATA BROWSER (BDPDB): A BROWSER-BASED TOOL FOR MANAGING, SEARCHING, AND VIEWING MRI DATA FROM PATIENTS WITH BRAIN LESIONS

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BDPDB allows researchers to easily search, browse, visualize, and explore large MRI datasets collected from patients with focal brain lesions.

BACKGROUND

Recent years have seen the growth of large, heterogeneous neuroimaging samples of neurological patients with focal brain lesions. Patients in these samples often differ in the causes, anatomical locations, and functional consequences of their lesions, and may participate in studies with different imaging and assessment protocols. When analyzing data from these samples, it is often necessary to select a sub-set of patients whose demographics, clinical characteristics, and patterns of damage are most appropriate for answering the question at hand. This selection process can be difficult and time consuming, particularly when researchers need to limit their analysis to patients with damage to a particular brain area or set of areas. We present an open source, web browser-based database application designed to aid researchers in the management, exploration, visualization and analysis of these complex datasets.

IMPLEMENTATION

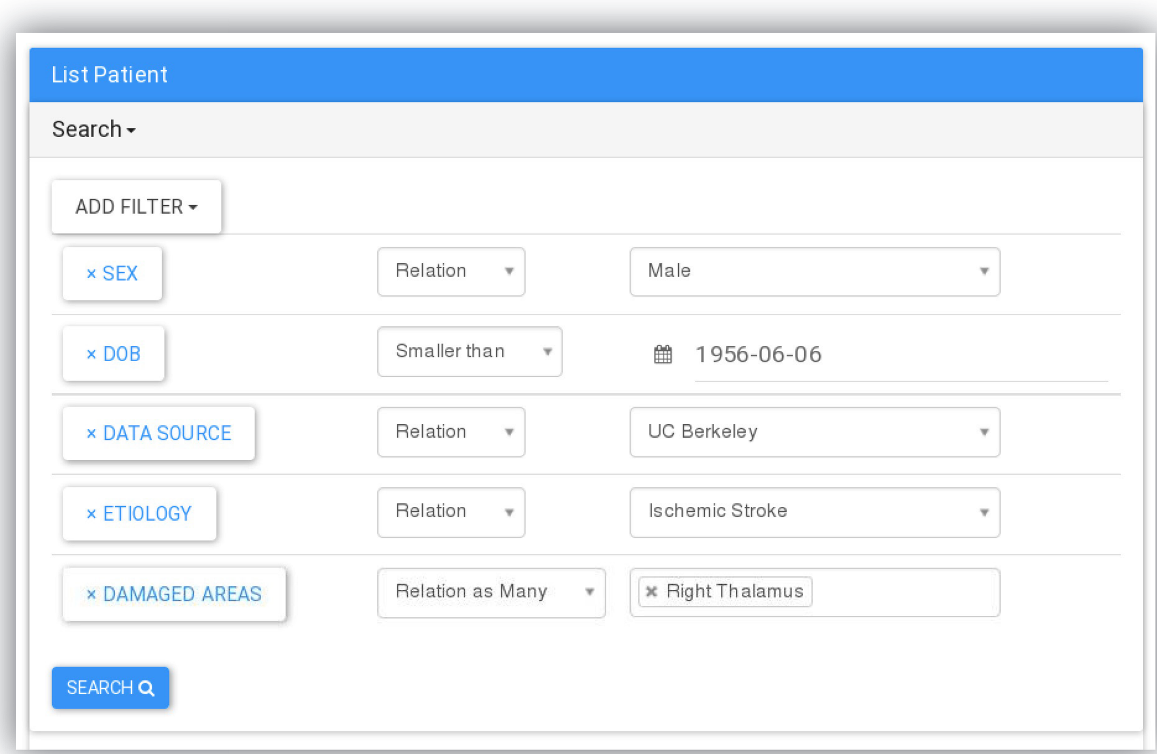
BDPDB is written in Python and can be installed and run on most modern Unix-like systems (e.g. Linux, OS X).

Patient demographic information (e.g. age, sex) and clinical characteristics (e.g. cause of damage, age of injury) are stored in an SQLite database and exposed to the user through the Flask, Jinja2, and SQLAlchemy (managed with the Flask-Appbuilder package).

Location-based search is accomplished by loading the MNI space binary lesion mask image from each patient into a common Pandas DataFrame (stored on disk as an HDF5 file) and indexing over voxels. By selectively loading only voxels relevant to the current search (i.e. those voxels at the specified MNI coordinate; 1 per patient), we are able to achieve near real-time search speeds (~10ms per query).

Visualization of the lesion mask overlap heat map, individual patient lesion masks and patient images is handled by the Papaya image viewer, which reads scan data directly from NIfTI files stored locally or served remotely.

Easily construct complex filter queries.



Get BDPDB at http://github.com/danlurie/bdpdb

Questions or comments are welcome: dan.lurie@berkeley.edu

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FEATURES

Search for patients by lesion location.

Specify MNI coordinates, select a location using the interactive lesion overlap heat map viewer, or pick a brain area from an atlas.

The lesion mask overlap heat map is automatically re-generated every time a patient mask is added or removed from the image database.

See all your patient data in one place.

View patient masks in native space or on the MNI template.

All patient metadata is searchable, and access can be limited to authorized users.

Click through to view scans without leaving the browser